



UNIVERSITY OF ASIA PACIFIC
Department of Computer Science and Engineering

PROJECT PROPOSAL
Smart Classroom Management and Automation System

Course Code: CSE 316

Course Name: Microprocessors and Microcontrollers Lab

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1. Project Overview

The Smart Classroom Management and Automation System is a microcontroller-based embedded system designed to make classroom management more efficient and automated. The system identifies teachers and registered students using RFID, detects entry and exit using IR sensors, monitors classroom temperature, and automatically controls classroom appliances based on classroom occupancy and environmental conditions.

An RFID reader is used to identify the teacher and registered students. Two IR sensors are placed at the classroom entrance to determine whether a person is entering or leaving. This allows the system to maintain the number of students present in the classroom. If an unregistered person enters without a valid RFID card, the system identifies them as an unknown person and does not add them to the student count.

A temperature sensor is used to monitor the classroom temperature. Based on teacher & student presence and temperature conditions, the microcontroller can automatically control devices such as the fan, light, AC, and projector using relay-based switching. For the project prototype, low-voltage devices can be used to demonstrate the control mechanism instead of directly connecting real classroom appliances.

The system also displays important classroom information such as teacher status, student count, temperature, and appliance status on an LCD display.

2. Components Required

Component	Purpose
ESP32	Main controller for processing sensor data and making decisions
RFID RC522 Reader	Identifies teachers and registered students
RFID Cards	Used for teacher and student identification
IR Sensors × 2	Detect entry & exit and determine movement direction
DHT11 or DHT22 Temperature Sensor	Measures classroom temperature
16×2 I2C LCD	Displays classroom information and system status
Relay Module	Controls connected appliances such as fan, light, AC, and projector
Small DC Fan	Prototype representation of classroom fan
LED	Prototype representation of classroom lighting
Buzzer	Provides alerts for unknown/unauthorized persons or other events
Breadboard	Used for circuit prototyping
Jumper Wires	Used for component connections
5V or USB Power Supply	Provides power to the prototype

3. Features

Teacher Identification

- Identifies the teacher using an RFID card.
- Displays the teacher's identification and status on the LCD.

Student Identification

- Registered students can be identified using RFID cards.
- The system distinguishes registered students from unknown persons.

Automatic Student Counting

- Two IR sensors detect the direction of movement.
- Entry increases the student count by one.
- Exit decreases the student count by one.

Unknown Person Detection

- A person without a registered RFID card is treated as an unknown person.
- Unknown persons are not included in the student count.
- The system can provide a buzzer or LCD alert.

Classroom Temperature Monitoring

- Continuously monitors the classroom temperature using a temperature sensor.
- Displays the current temperature on the LCD.

Automatic Fan Control

- Controls the fan according to classroom occupancy and temperature conditions.

Automatic Light Control

- Automatically turns the classroom light ON when the classroom is occupied.
- Turn the light OFF when the classroom becomes empty.

Automatic AC Control

- Controls the AC according to classroom occupancy and temperature conditions.

Automatic Projector Control

- Allows the projector to be automatically controlled based on teacher or classroom status.

Real-Time Classroom Status Display

- Displays information such as: teacher status, student count, temperature, fan status, light status, AC status, and projector status.

Automated Classroom Management

- The system combines identification, counting, environmental monitoring, decision-making, and appliance control into a single smart classroom system.

4. Mobile App Integration

The system includes a mobile application connected to the ESP32 over Wi-Fi. All the classroom information shown on the LCD is also sent to the app and stored there, so authorized users can view and review it at any time.

Authorized Access

- Only authorized users (teacher/admin) can log in and view the data.

Real-Time Classroom Information

- Teacher status
- Student count
- Classroom temperature
- Fan, light, AC, and projector status

Unknown Person Alert

- Unknown person detection is shown in the app with date and time.

Data Storage

- All the information above is stored in the app with timestamps, so previous records can be viewed later.
- Firebase for data storing

Additional Components Required

Component	Purpose
Mobile Application	Displays and stores classroom information for authorized users